2.5. GAUSSIAN ELIMINATION METHOD

There are a lot of known Gaussian elimination method schemes intended to find a solution to the system of linear algebraic equations $Ax = b$, which are adapted for manual or machine calculation of general or special type matrices, where $A \equiv \left[a_{ij} \right]$ is the square $n \times n$ matrix of coefficients with unknowns; $x \equiv \left[x_i \right]$ is the column vector of the unknowns; $b \equiv \left[b_i \right]$ is the column vector of the right-hand sides of the system.

So, the following SLAE is given:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n = b_2 \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ a_{n1}x_1 + a_{n2}x_2 + \ldots + a_{nn}x_n = b_n \end{cases}.$$

The Gaussian method can be interpreted as a method in which initially the matrix A is reduced by means of equivalent transformations for the SLAE (rows exchange, multiplication of a row by a number, addition of rows and replacement of one of them with the result) to the upper triangular form (forward path) and then to the unity form (backward path). The unknowns are determined during the backward path: obviously, if the SLAE matrix is a unity one, then $x_i = b_i$, $i = 1,\ldots,n$.

Let's write the extended matrix of the system:

$$\begin{array}{l} \\ \textit{Leading row} \rightarrow \end{array} \left[\begin{array}{c|cccccc} & x_1 & x_2 & x_3 & \cdots & x_n & b \\ \hline & a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ & a_{21} & a_{22} & a_{23} & \cdots & a_{2n} & b_2 \\ & a_{31} & a_{32} & a_{33} & \cdots & a_{3n} & b_3 \\ & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ & a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} & b_n \end{array} \right] \begin{array}{l} \left(-a_{21}/a_{11}\right);\left(-a_{31}/a_{11}\right);\ldots;\left(-a_{n1}/a_{11}\right) \\ \\ \xrightarrow[1st\ \ step]{} \\ \\ \uparrow \textit{Leading column} \end{array}.$$

At the first step of the Gaussian algorithm, we should choose the diagonal element $a_{11} \neq 0$ (if it is equal to zero, then we must replace the first row with any underlying row, where $a_{i1} \neq 0$, $i = 2,\ldots,n$) and declare it the *leading element*; the corresponding row and column at the intersection of which it stands will be the *leading row and column*. Let's zero out the elements $a_{21},\ldots,a_{n1}$ of the leading

column. For this purpose, we should form the numbers $(-a_{21}/a_{11});(-a_{31}/a_{11});\ldots;(-a_{n1}/a_{11})$, which we shall correlate with the leading row. Multiplying the leading row by the number $(-a_{21}/a_{11})$, adding it to the second row, and putting the result in place of the second row, we shall get zero instead of the element a_{21} and the elements $a_{2j}^{1}=a_{2j}+a_{1j}(-a_{21}/a_{11})$, $j=2,\ldots,n$, $b_2^1=b_2+b_1(-a_{21}/a_{11})$ instead of the elements a_{2j}, $j=2,\ldots,n$, b_2. We continue so up to the last row: multiplying the leading row by the number $(-a_{n1}/a_{11})$, adding it to the *n*-th row, and putting the result in place of the *n*-th row, we shall get zero instead of the element a_{n1}, and the remaining elements of this row will be as follows: $a_{nj}^{1}=a_{nj}+a_{1j}(-a_{n1}/a_{11})$, $b_n^1=b_n+b_1(-a_{n1}/a_{11})$.

Keeping the leading first row of the matrix unchanged at the first step, we shall obtain the following matrix as a result of the first step of the Gaussian algorithm:

$$\begin{array}{c} \\ \textit{Leading row}\rightarrow \\ \\ \\ \\ \\ \uparrow \textit{ Leading column} \end{array} \begin{array}{c} \begin{array}{cccccc} x_1 & x_2 & x_3 & \cdots & x_n & b \end{array} \\ \left[\begin{array}{c|c|ccc|c} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ \hline 0 & a_{22}^1 & a_{23}^1 & \cdots & a_{2n}^1 & b_2^1 \\ \hline 0 & a_{32}^1 & a_{33}^1 & \cdots & a_{3n}^1 & b_3^1 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & a_{n2}^1 & a_{n3}^1 & \cdots & a_{nn}^1 & b_n^1 \end{array}\right] \end{array} \begin{array}{l} \left(-a_{32}^1/a_{22}^1\right);\ldots;\left(-a_{n2}^1/a_{22}^1\right) \\ \xrightarrow[2nd\ \ step]{} \end{array} .$$

At the second step of the Gaussian algorithm, the element $a_{22}^1\neq 0$ is selected as the leading element (if it is equal to zero, then the second row is mutually replaced with the underlying row, where $a_{i2}^1\neq 0$, $i=3,\ldots,n$). The numbers $\left(-\frac{a_{32}^1}{a_{22}^1}\right);\ldots;\left(-\frac{a_{n2}^1}{a_{22}^1}\right)$ are formed, which are placed near the leading row. Multiplying the leading row by the number $\left(-\frac{a_{32}^1}{a_{22}^1}\right)$ and adding the result to the third row, we shall get zero instead of the element a_{32}^1 and the elements $a_{3j}^2=a_{3j}^1+a_{2j}^1\left(-\frac{a_{32}^1}{a_{22}^1}\right)$,

$j=3\ldots n$, $b_3^2=b_3^1+b_2^1\left(-\frac{a_{32}^1}{a_{22}^1}\right)$ instead of the elements a_{3j}^1, $j=3,\ldots,n$, b_3^1. And so on up to the last row: multiplying the leading row by the number $\left(-\frac{a_{n2}^1}{a_{22}^1}\right)$, adding the result to the *n*-th row, and putting the resulting sum in the place of the *n*-th row, we shall get zero instead of the element a_{n2}^1, and the elements $a_{nj}^2=a_{nj}^1+a_{2j}^1\left(-\frac{a_{n2}^1}{a_{22}^1}\right)$, $j=3,\ldots,n$, $b_n^2=b_n^1+b_2^1\left(-\frac{a_{n2}^1}{a_{22}^1}\right)$ instead of the elements a_{nj}^1, b_n^1.

Keeping the first and leading second rows of the matrix unchanged at the second step, we shall obtain the following matrix as a result of the second step of the Gaussian algorithm:

$$\begin{array}{l} \\ \textit{Leading row}\rightarrow \\ \\ \\ \\ \\ \\ \\ \end{array}\begin{array}{c} \begin{array}{cccccc} x_1 & x_2 & x_3 & \cdots & x_n & b \end{array} \\ \left[\begin{array}{cc|c|ccc} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ 0 & a_{22}^1 & a_{23}^1 & \cdots & a_{2n}^1 & b_2^1 \\ \hline 0 & 0 & a_{33}^2 & \cdots & a_{3n}^2 & b_3^2 \\ \hline \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & a_{n3}^2 & \cdots & a_{nn}^2 & b_n^2 \end{array}\right] \\ \uparrow\ \textit{Leading column} \end{array} \xrightarrow{3rd\ \ step}\ldots\xrightarrow{(n-1)-th\ \ step}.$$

As a result, after the $(n-1)$-th step of the Gaussian algorithm, we shall obtain the following extended matrix containing the upper triangular matrix of the SLAE:

$$\begin{array}{c} \begin{array}{cccccc} x_1 & x_2 & x_3 & \ldots & x_n & b \end{array} \\ \left[\begin{array}{cccccc} a_{11} & a_{12} & a_{13} & \ldots & a_{1n} & b_1 \\ 0 & a_{22}^1 & a_{23}^1 & \ldots & a_{2n}^1 & b_2^1 \\ 0 & 0 & a_{33}^2 & \ldots & a_{3n}^2 & b_3^2 \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ 0 & 0 & 0 & \ldots & a_{nn}^{n-1} & b_n^{n-1} \end{array}\right] \end{array}.$$

The forward path of the Gaussian algorithm is completed.

If the elements of any row of the matrix of the system become equal to zero as a result of transformations and the right-hand side is not equal to zero, then the SLAE is incompatible and has no solution. If the elements of any row of the matrix of the system and the right-hand side become equal to zero as a result of

transformations, then the SLAE is compatible but has an infinite number of solutions.

The determinant of the triangular matrix is equal to the product of the diagonal elements. As a result of the forward path of the Gaussian elimination method, the system of linear equations is reduced to the upper triangular matrix. Therefore, as a result of the forward path of the Gaussian method, we can immediately calculate the determinant of the matrix A of the original SLAE as the product of the leading elements:

$$\det A = (-1)^p a_{11} a_{22}^1 a_{33}^2 \cdot \ldots \cdot a_{nn}^{n-1},$$

where p is the amount of row permutations during the forward path.

During the backward path of the Gaussian algorithm, the extended matrix of the system obtained after the forward path can be reduced to the diagonal matrix by similar equivalent transformations (if we also turn to zero all coefficients located above the main diagonal) and then (if we divide each equation by the corresponding element located on the main diagonal) to the unity matrix, which will allow us to obtain the solution to the original SLAE.

However, a more efficient technique is to write down the solution to the system sequentially through recurrence relations, when x_n is immediately determined from the last equation, x_{n-1} – from the second-to-last one and so on, until x_1 is determined from the first equation:

$$\begin{cases} a_{nn}^{n-1} x_n = b_n^{n-1} & \Rightarrow x_n \\ a_{n-1n-1}^{n-2} x_{n-1} + a_{n-1n}^{n-2} x_n = b_{n-1}^{n-2} & \Rightarrow x_{n-1} \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots & \\ a_{11} x_1 + \ldots + a_{1n} x_n = b_1 & \Rightarrow x_1 \end{cases}, \quad \begin{cases} x_n = \dfrac{b_n^{n-1}}{a_{nn}^{n-1}} \\ x_i = \dfrac{1}{a_{ii}^{i-1}} \left[b_i^{i-1} - \sum\limits_{k=i+1}^{n} a_{ik}^{i-1} x_k \right], i = n-1, \ldots, 1 \end{cases}.$$

It is important to note that the implementation of the Gaussian method requires around n^3 arithmetic operations, which makes this method *expensive*. Moreover, the number of operations is determined mainly by the operations carried out when performing the forward path of the Gaussian method. The backward path of the Gaussian method (through recurrent relations) requires around n^2 operations. Therefore, if it is required to solve several systems of linear algebraic

equations $Ax = b$ with the same matrix and different right-hand sides, then, in this case, it is advisable to implement the Gaussian method algorithm in the form of two subprograms: the first subprogram must implement the forward path of the algorithm and obtain the upper triangular matrix of the SLAE at the output, and the second subprogram must implement the forward path of the algorithm for the right-hand sides and, using the resulting upper triangular matrix, calculate the solution to the system for the specific right-hand side.

2.6. MATRIX INVERSION USING GAUSSIAN METHOD

The Gaussian elimination method can also be applied to invert a nondegenerate ($\det A \neq 0$) matrix. Indeed, suppose it is required to invert the nondegenerate matrix *A*. Then, having made the notation $A^{-1} = X$, we can write the matrix equation $AX = E$, where *E* is the unity matrix, on the basis of which we can write the chain of SLAEs:

$$A \cdot \begin{pmatrix} x_{11} \\ x_{21} \\ \dots \\ x_{n1} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ \dots \\ 0 \end{pmatrix} = e_1, \quad A \cdot \begin{pmatrix} x_{12} \\ x_{22} \\ \dots \\ x_{n2} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ \dots \\ 0 \end{pmatrix} = e_2, \ \dots, \quad A \cdot \begin{pmatrix} x_{1n} \\ x_{2n} \\ \dots \\ x_{nn} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \dots \\ 1 \end{pmatrix} = e_n,$$

each of which can be solved using the Gaussian method.

So, for matrix inversion, it is necessary to solve *n* systems of linear equations of the *n*-th order with the same matrix *A* and different right-hand sides. Since the upper triangular matrix for all these SLAEs is the same, the forward path of the Gaussian method is applied for the matrix *A* just once. The following extended matrix is constructed:

$$\left.\begin{array}{cccc|cccc} x_{1n} & x_{2n} & \dots & x_{nn} & & & & \\ \dots & \dots & \dots & \dots & & & & \\ x_{12} & x_{22} & \dots & x_{n2} & & & & \\ x_{11} & x_{21} & \dots & x_{n1} & b^1 & b^2 & \dots & b^n \\ \hline a_{11} & a_{12} & \dots & a_{1n} & 1 & 0 & \dots & 0 \\ a_{21} & a_{22} & \dots & a_{2n} & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} & 0 & 0 & \dots & 1 \end{array}\right|.$$

As a result of applying $(n-1)$ steps of the Gaussian method, we obtain:

$$\left.\begin{array}{cccc|cccc} x_{1n} & x_{2n} & \dots & x_{nn} & & & & \\ \dots & \dots & \dots & \dots & & & & \\ x_{12} & x_{22} & \dots & x_{n2} & & & & \\ x_{11} & x_{21} & \dots & x_{n1} & b^1 & b^2 & \dots & b^n \\ \hline a_{11} & a_{12} & \dots & a_{1n} & b_{11} & b_{12} & \dots & b_{1n} \\ 0 & a_{22}^1 & \dots & a_{2n}^1 & b_{21}^1 & b_{22}^1 & \dots & b_{2n}^1 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & a_{nn}^{n-1} & b_{n1}^{n-1} & b_{n2}^{n-1} & \dots & b_{nn}^{n-1} \end{array}\right| .$$

Herewith, the first column $(x_{11} \;\; x_{21} \;\; \dots \;\; x_{n1})^T$ of the inverse matrix is determined during the backward path of the Gaussian method with the right-hand side b^1, the second column $(x_{12} \;\; x_{22} \;\; \dots \;\; x_{n2})^T$ – with the right-hand side b^2, and so on. The column $(x_{1n} \;\; x_{2n} \;\; \dots \;\; x_{nn})^T$ is determined with the right-hand side b^n.

However, as an alternative, during the backward path of the Gaussian method it is possible to reduce the whole matrix A from the upper triangular matrix directly to the unity matrix through equivalent transformations in order to immediately obtain its inverse matrix, this time without losing algorithmical efficiency.

Example. Solve the SLAE using the Gaussian method, calculate the determinant and the inverse matrix for the matrix of the SLAE:

$$\begin{cases} x_1 + 3x_2 - x_3 + 2x_4 = 13 \\ 6x_1 - 2x_2 + 2x_4 = 20 \\ 3x_1 - 5x_2 + x_3 + 8x_4 = 7 \\ -x_1 + 4x_2 - 5x_3 + 9x_4 = 7 \end{cases} ; \; A = \begin{pmatrix} 1 & 3 & -1 & 2 \\ 6 & -2 & 0 & 2 \\ 3 & -5 & 1 & 8 \\ -1 & 4 & -5 & 9 \end{pmatrix}, \; b = \begin{pmatrix} 13 \\ 20 \\ 7 \\ 7 \end{pmatrix}.$$

Forward path:

$$\begin{array}{cccc} x_1 & x_2 & x_3 & x_4 \end{array} \; b$$

$$\left(\begin{array}{cccc|c} \mathbf{1} & 3 & -1 & 2 & 13 \\ 6 & -2 & 0 & 2 & 20 \\ 3 & -5 & 1 & 8 & 7 \\ -1 & 4 & -5 & 9 & 7 \end{array}\right) \begin{array}{l} (-6/1); (-3/1); (1/1); \\ \\ \\ \\ \end{array} \xrightarrow[1st \;\; step]{}$$

$$\begin{matrix} x_1 & x_2 & x_3 & x_4 & b \end{matrix}$$
$$\left(\begin{array}{cccc|c} 1 & 3 & -1 & 2 & 13 \\ 0 & \mathbf{-20} & 6 & -10 & -58 \\ 0 & -14 & 4 & 2 & -32 \\ 0 & 7 & -6 & 11 & 20 \end{array}\right) (-14/20);(7/20); \xrightarrow[2nd\ \ step]{}$$

$$\begin{matrix} x_1 & x_2 & x_3 & x_4 & b \end{matrix}$$
$$\left(\begin{array}{cccc|c} 1 & 3 & -1 & 2 & 13 \\ 0 & -20 & 6 & -10 & -58 \\ 0 & 0 & \mathbf{-1/5} & 9 & 43/5 \\ 0 & 0 & -39/10 & 15/2 & -3/10 \end{array}\right) (-39/2); \xrightarrow[3rd\ \ step]{}$$

$$\begin{matrix} x_1 & x_2 & x_3 & x_4 & b \end{matrix}$$
$$\left(\begin{array}{cccc|c} 1 & 3 & -1 & 2 & 13 \\ 0 & -20 & 6 & -10 & -58 \\ 0 & 0 & -1/5 & 9 & 43/5 \\ 0 & 0 & 0 & -168 & -168 \end{array}\right).$$

The determinant of the matrix: $\det A = 1\cdot(-20)\cdot(-1/5)\cdot(-168) = -672$.

Backward path:

$$\begin{cases} -168x_4 = -168 \\ (-1/5)x_3 + 9x_4 = 43/5 \\ -20x_2 + 6x_3 - 10x_4 = -58 \\ x_1 + 3x_2 - x_3 + 2x_4 = 13 \end{cases}; \begin{cases} x_4 = 1 \\ (-1/5)x_3 = 43/5 - 9\cdot 1 = -2/5,\ x_3 = 2 \\ -20x_2 = -58 - 6\cdot 2 + 10\cdot 1 = -60,\ x_2 = 3 \\ x_1 = 13 - 3\cdot 3 + 2 - 2\cdot 1 = 4,\ x_1 = 4 \end{cases}.$$

Solution to the SLAE: $x = \begin{pmatrix} 4 & 3 & 2 & 1 \end{pmatrix}^T$.

Finding the inverse matrix A^{-1}.

Forward path:

$$\begin{matrix} x_{14} & x_{24} & x_{34} & x_{44} \\ x_{13} & x_{23} & x_{33} & x_{43} \\ x_{12} & x_{22} & x_{32} & x_{42} \\ x_{11} & x_{21} & x_{31} & x_{41} & b^1 & b^2 & b^3 & b^4 \end{matrix}$$
$$\left(\begin{array}{cccc|cccc} \mathbf{1} & 3 & -1 & 2 & 1 & 0 & 0 & 0 \\ 6 & -2 & 0 & 2 & 0 & 1 & 0 & 0 \\ 3 & -5 & 1 & 8 & 0 & 0 & 1 & 0 \\ -1 & 4 & -5 & 9 & 0 & 0 & 0 & 1 \end{array}\right) (-6/1);(-3/1);(1/1); \xrightarrow[1st\ \ step]{}$$

$$
\begin{array}{cccc}
x_{14} & x_{24} & x_{34} & x_{44} \\
x_{13} & x_{23} & x_{33} & x_{43} \\
x_{12} & x_{22} & x_{32} & x_{42} \\
x_{11} & x_{21} & x_{31} & x_{41}
\end{array}
\quad
\begin{array}{cccc}
b^1 & b^2 & b^3 & b^4
\end{array}
$$

$$
\left(\begin{array}{cccc|cccc}
1 & 3 & -1 & 2 & 1 & 0 & 0 & 0 \\
0 & \mathbf{-20} & 6 & -10 & -6 & 1 & 0 & 0 \\
0 & -14 & 4 & 2 & -3 & 0 & 1 & 0 \\
0 & 7 & -6 & 11 & 1 & 0 & 0 & 1
\end{array}\right)
(-14/20);(7/20);
\xrightarrow[2nd\ \ step]{}
$$

$$
\begin{array}{cccc}
x_{14} & x_{24} & x_{34} & x_{44} \\
x_{13} & x_{23} & x_{33} & x_{43} \\
x_{12} & x_{22} & x_{32} & x_{42} \\
x_{11} & x_{21} & x_{31} & x_{41}
\end{array}
\quad
\begin{array}{cccc}
b^1 & b^2 & b^3 & b^4
\end{array}
$$

$$
\left(\begin{array}{cccc|cccc}
1 & 3 & -1 & 2 & 1 & 0 & 0 & 0 \\
0 & -20 & 6 & -10 & -6 & 1 & 0 & 0 \\
0 & 0 & \mathbf{-1/5} & 9 & 6/5 & -7/10 & 1 & 0 \\
0 & 0 & -39/10 & 15/2 & -11/10 & 7/20 & 0 & 1
\end{array}\right)
(-39/2);
\xrightarrow[3rd\ \ step]{}
$$

$$
\begin{array}{cccc}
x_{14} & x_{24} & x_{34} & x_{44} \\
x_{13} & x_{23} & x_{33} & x_{43} \\
x_{12} & x_{22} & x_{32} & x_{42} \\
x_{11} & x_{21} & x_{31} & x_{41}
\end{array}
\quad
\begin{array}{cccc}
b^1 & b^2 & b^3 & b^4
\end{array}
$$

$$
\left(\begin{array}{cccc|cccc}
1 & 3 & -1 & 2 & 1 & 0 & 0 & 0 \\
0 & -20 & 6 & -10 & -6 & 1 & 0 & 0 \\
0 & 0 & -1/5 & 9 & 6/5 & -7/10 & 1 & 0 \\
0 & 0 & 0 & -168 & -49/2 & 14 & -39/2 & 1
\end{array}\right).
$$

Backward path:

$$
\begin{cases}
-168x_{41} = -49/2 \\
(-1/5)x_{31} + 9x_{41} = 6/5 \\
-20x_{21} + 6x_{31} - 10x_{41} = -6 \\
x_{11} + 3x_{21} - x_{31} + 2x_{41} = 1
\end{cases};
\begin{cases}
-168x_{42} = 14 \\
(-1/5)x_{32} + 9x_{42} = -7/10 \\
-20x_{22} + 6x_{32} - 10x_{42} = 1 \\
x_{12} + 3x_{22} - x_{32} + 2x_{42} = 0
\end{cases};
$$

$$
\begin{cases}
-168x_{43} = -39/2 \\
(-1/5)x_{33} + 9x_{43} = 1 \\
-20x_{23} + 6x_{33} - 10x_{43} = 0 \\
x_{13} + 3x_{23} - x_{33} + 2x_{43} = 0
\end{cases};
\begin{cases}
-168x_{44} = 1 \\
(-1/5)x_{34} + 9x_{44} = 0 \\
-20x_{24} + 6x_{34} - 10x_{44} = 0 \\
x_{14} + 3x_{24} - x_{34} + 2x_{44} = 0
\end{cases}.
$$

From here $A^{-1}=\begin{pmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{pmatrix}=\begin{pmatrix} 1/12 & 1/6 & -1/28 & -1/42 \\ 19/48 & -1/12 & 1/112 & -13/168 \\ 9/16 & -1/4 & 25/112 & -15/56 \\ 7/48 & -1/12 & 13/112 & -1/168 \end{pmatrix}$.

2.7. GAUSSIAN METHOD WITH PIVOT ELEMENT SELECTION

At some *i*-th step of the forward path of the Gaussian elimination method, it may turn out that the coefficient $a_{ii}^{i-1}\neq 0$, but it is small in modulus in comparison to the remaining elements of the system matrix and, in particular, in comparison to the elements of the *i*-th column below. Division of the system coefficients by a small quantity can lead to significant rounding errors or even the overflow of variables.

To reduce these errors, proceed as follows. Select the largest in absolute value element (*pivot element*) among the elements of the *i*-th column a_{ki}^{i-1}, $k=i,\ldots,n$ of each intermediate matrix and make the pivot element the leading element by rearranging the *i*-th row and the row containing the pivot element. Such a modification of the Gaussian elimination method is called the Gaussian method with pivot element selection. The case of zero leading elements emergence in this goes by itself.

Example. Solve the SLAE using the Gaussian method with pivot element selection, calculate the determinant and the inverse matrix for the matrix of the SLAE:

$$\begin{cases} x_1+3x_2-x_3+2x_4=13 \\ 6x_1-2x_2+2x_4=20 \\ 3x_1-5x_2+x_3+8x_4=7 \\ -x_1+4x_2-5x_3+9x_4=7 \end{cases};\ A=\begin{pmatrix} 1 & 3 & -1 & 2 \\ 6 & -2 & 0 & 2 \\ 3 & -5 & 1 & 8 \\ -1 & 4 & -5 & 9 \end{pmatrix},\ b=\begin{pmatrix} 13 \\ 20 \\ 7 \\ 7 \end{pmatrix}.$$

Forward path:

$$\begin{array}{cccc|c} x_1 & x_2 & x_3 & x_4 & b \\ \hline 1 & 3 & -1 & 2 & 13 \\ \mathbf{6} & -2 & 0 & 2 & 20 \\ 3 & -5 & 1 & 8 & 7 \\ -1 & 4 & -5 & 9 & 7 \end{array} \xrightarrow{1st\ \ step} \begin{array}{cccc|c} x_1 & x_2 & x_3 & x_4 & b \\ \hline 6 & -2 & 0 & 2 & 20 \\ 0 & 10/3 & -1 & 5/3 & 29/3 \\ 0 & \mathbf{-4} & 1 & 7 & -3 \\ 0 & 11/3 & -5 & 28/3 & 31/3 \end{array} \xrightarrow{2nd\ \ step}$$

$$\begin{array}{cccc|c} x_1 & x_2 & x_3 & x_4 & b \\ \left(\begin{array}{cccc|c} 6 & -2 & 0 & 2 & 20 \\ 0 & -4 & 1 & 7 & -3 \\ 0 & 0 & -1/6 & 15/2 & 43/6 \\ 0 & 0 & \mathbf{-49/12} & 63/4 & 91/12 \end{array}\right) & \xrightarrow{3rd\ \ step} & \left(\begin{array}{cccc|c} 6 & -2 & 0 & 2 & 20 \\ 0 & -4 & 1 & 7 & -3 \\ 0 & 0 & -49/12 & 63/4 & 91/12 \\ 0 & 0 & 0 & 48/7 & 48/7 \end{array}\right) \end{array}.$$

The amount of permutations performed in 3 steps: $p=3$.

The determinant of the matrix: $\det A=(-1)^3\cdot 6\cdot(-4)\cdot(-49/12)\cdot(48/7)=-672$.

Backward path:

$$\begin{cases} (48/7)x_4=48/7 \\ (-49/12)x_3+(63/4)x_4=91/12 \\ -4x_2+x_3+7x_4=-3 \\ 6x_1-2x_2+2x_4=20 \end{cases};\quad \begin{cases} x_4=1 \\ (-49/12)x_3=91/12-(63/4)\cdot 1=-98/12,\ x_3=2 \\ -4x_2=-3-2-7\cdot 1=-12,\ x_2=3 \\ 6x_1=20+2\cdot 3-2\cdot 1=24,\ x_1=4 \end{cases}.$$

Solution to the SLAE: $x=\begin{pmatrix}4 & 3 & 2 & 1\end{pmatrix}^T$.

Finding the inverse matrix A^{-1}.

Forward path:

$$\begin{array}{cccc|cccc} x_{14} & x_{24} & x_{34} & x_{44} & & & & \\ x_{13} & x_{23} & x_{33} & x_{43} & & & & \\ x_{12} & x_{22} & x_{32} & x_{42} & & & & \\ x_{11} & x_{21} & x_{31} & x_{41} & b^1 & b^2 & b^3 & b^4 \\ \left(\begin{array}{cccc|cccc} 1 & 3 & -1 & 2 & 1 & 0 & 0 & 0 \\ \mathbf{6} & -2 & 0 & 2 & 0 & 1 & 0 & 0 \\ 3 & -5 & 1 & 8 & 0 & 0 & 1 & 0 \\ -1 & 4 & -5 & 9 & 0 & 0 & 0 & 1 \end{array}\right) & \xrightarrow{1st\ \ step} & & & & & & \end{array}$$

$$\begin{array}{cccc|cccc} x_{14} & x_{24} & x_{34} & x_{44} & & & & \\ x_{13} & x_{23} & x_{33} & x_{43} & & & & \\ x_{12} & x_{22} & x_{32} & x_{42} & & & & \\ x_{11} & x_{21} & x_{31} & x_{41} & b^1 & b^2 & b^3 & b^4 \\ \left(\begin{array}{cccc|cccc} 6 & -2 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 10/3 & -1 & 5/3 & 1 & -1/6 & 0 & 0 \\ 0 & \mathbf{-4} & 1 & 7 & 0 & -1/2 & 1 & 0 \\ 0 & 11/3 & -5 & 28/3 & 0 & 1/6 & 0 & 1 \end{array}\right) & \xrightarrow{2nd\ \ step} & & & & & & \end{array}$$

$$\begin{array}{cccc|cccc} x_{14} & x_{24} & x_{34} & x_{44} & & & & \\ x_{13} & x_{23} & x_{33} & x_{43} & & & & \\ x_{12} & x_{22} & x_{32} & x_{42} & & & & \\ x_{11} & x_{21} & x_{31} & x_{41} & b^1 & b^2 & b^3 & b^4 \\ \hline 6 & -2 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & -4 & 1 & 7 & 0 & -1/2 & 1 & 0 \\ 0 & 0 & -1/6 & 15/2 & 1 & -7/12 & 5/6 & 0 \\ 0 & 0 & \mathbf{-49/12} & 63/4 & 0 & -7/24 & 11/12 & 1 \end{array} \xrightarrow{3rd\ \ step}$$

$$\begin{array}{cccc|cccc} x_{14} & x_{24} & x_{34} & x_{44} & & & & \\ x_{13} & x_{23} & x_{33} & x_{43} & & & & \\ x_{12} & x_{22} & x_{32} & x_{42} & & & & \\ x_{11} & x_{21} & x_{31} & x_{41} & b^1 & b^2 & b^3 & b^4 \\ \hline 6 & -2 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & -4 & 1 & 7 & 0 & -1/2 & 1 & 0 \\ 0 & 0 & -49/12 & 63/4 & 0 & -7/24 & 11/12 & 1 \\ 0 & 0 & 0 & 48/7 & 1 & -4/7 & 39/49 & -2/49 \end{array}.$$

Backward path:

$$\begin{cases} (48/7)x_{41} = 1 \\ (-49/12)x_{31} + (63/4)x_{41} = 0 \\ -4x_{21} + x_{31} + 7x_{41} = 0 \\ 6x_{11} - 2x_{21} + 2x_{41} = 0 \end{cases}; \begin{cases} (48/7)x_{42} = -4/7 \\ (-49/12)x_{32} + (63/4)x_{42} = -7/24 \\ -4x_{22} + x_{32} + 7x_{42} = -1/2 \\ 6x_{12} - 2x_{22} + 2x_{42} = 1 \end{cases};$$

$$\begin{cases} (48/7)x_{43} = 39/49 \\ (-49/12)x_{33} + (63/4)x_{43} = 11/12 \\ -4x_{23} + x_{33} + 7x_{43} = 1 \\ 6x_{13} - 2x_{23} + 2x_{43} = 0 \end{cases}; \begin{cases} (48/7)x_{44} = -2/49 \\ (-49/12)x_{34} + (63/4)x_{44} = 1 \\ -4x_{24} + x_{34} + 7x_{44} = 0 \\ 6x_{14} - 2x_{24} + 2x_{44} = 0 \end{cases}.$$

From here $A^{-1} = \begin{pmatrix} x_{11} & x_{12} & x_{13} & x_{14} \\ x_{21} & x_{22} & x_{23} & x_{24} \\ x_{31} & x_{32} & x_{33} & x_{34} \\ x_{41} & x_{42} & x_{43} & x_{44} \end{pmatrix} = \begin{pmatrix} 1/12 & 1/6 & -1/28 & -1/42 \\ 19/48 & -1/12 & 1/112 & -13/168 \\ 9/16 & -1/4 & 25/112 & -15/56 \\ 7/48 & -1/12 & 13/112 & -1/168 \end{pmatrix}.$

2.8. PROGRAM #03

Below is a proposed variant of the program algorithm for solving a SLAE, calculating the determinant and the inverse matrix of the SLAE's matrix using the Gaussian elimination method or the Gaussian method with pivot element selection.

```
ALGORITHM “Gaussian method with pivot element selection”
INPUT       n, a[n][n], b[n], l
OUTPUT      i, j, k, m, p, f, g, h, x[n], y, z[n][n]
BEGIN
    CYCLE “Rows” FOR i FROM 1 TO n BY 1
        CYCLE “Columns” FOR j FROM 1 TO n BY 1
            IF (i=j) z[i][j]:=1 ELSE z[i][j]:=0
    p:=0
    CYCLE “Forward path” FOR k FROM 1 TO n-1 BY 1
        #Part for the Gaussian method with pivot element selection
        IF (l≠0)
            IF (a[k][k]>0) g:=a[k][k] ELSE g:=-a[k][k]
            m:=k
            CYCLE “Rows” FOR i FROM k+1 TO n BY 1
                IF (a[i][k]>0) f:=a[i][k] ELSE f:=-a[i][k]
                IF (f>g) g:=f, m:=i
            IF (m≠k)
                CYCLE “Columns” FOR j FROM k TO n BY 1
                    g:=a[k][j], a[k][j]:=a[m][j], a[m][j]:=g
                g:=b[k], b[k]:=b[m], b[m]:=g
                CYCLE “Columns” FOR j FROM 1 TO n BY 1
                    g:=z[k][j], z[k][j]:=z[m][j], z[m][j]:=g
                p:=p+1
        #General part of the Gaussian method
        IF (a[k][k]=0) END
        CYCLE “Rows” FOR i FROM k+1 TO n BY 1
            h:=-a[i][k]/a[k][k]
            CYCLE “Columns” FOR j FROM k TO n BY 1
                a[i][j]:=a[i][j]+h*a[k][j]
            b[i]:=b[i]+h*b[k]
```

```
                    CYCLE “Columns” FOR j FROM 1 TO n BY 1
                        z[i][j]:=z[i][j]+h*z[k][j]
    y:=1-2*REMAINDER(p,2)
    CYCLE “Backward path” FOR k FROM n TO 1 BY -1
        y:=y*a[k][k]
        h:=0
        CYCLE “Rows” FOR i FROM k+1 TO n BY 1
            h:=h+a[k][i]*x[i]
        x[k]:=(b[k]-h)/a[k][k]
        CYCLE “Columns” FOR j FROM 1 TO n BY 1
            h:=0
            CYCLE “Rows” FOR i FROM k+1 TO n BY 1
                h:=h+a[k][i]*z[i][j]
            z[k][j]:=(z[k][j]-h)/a[k][k]
    PRINT x, y, z
END
```

2.9. *LU*-DECOMPOSITION OF MATRICES

The Gaussian method is often implemented on a computer with the help of *LU*-decomposition of matrices.

LU-decomposition of the matrix A is the matrix A decomposition into the product of the lower and upper triangular matrices:

$$A = LU,$$

where L is the lower triangular matrix ($l_{ij} = 0$, if $i < j$), and U is the upper triangular matrix ($u_{ij} = 0$, if $i > j$). Obviously, such a decomposition can be performed with an infinite number of ways; therefore, for its uniqueness, let's suppose that $l_{ii} = 1$, $i = 1,\ldots,n$.

The desired *LU*-decomposition can be constructed using the Gaussian method described above. Let's consider the k-th step of the Gaussian method, at

which the subdiagonal elements of the *k*-th column of the matrix $A^{(k-1)}$ are zeroed out. As described above, the following operation is used for this purpose:

$$a_{ij}^{(k)} = a_{ij}^{(k-1)} - \mu_i^{(k)} a_{kj}^{(k-1)}, \text{ where } \mu_i^{(k)} = \frac{a_{ik}^{(k-1)}}{a_{kk}^{(k-1)}},\ i = k+1,\ldots,n,\ j = k,\ldots,n\ \left(a_{kk}^{(k-1)} \neq 0\right).$$

In terms of matrix operations, such an operation is equivalent to performing multiplication $A^{(k)} = M_k A^{(k-1)}$, where elements of the matrix M_k are defined as follows:

$$m_{ij}^k = \begin{cases} 1, & i = j \\ -\mu_i^{(k)}, & j = k,\quad i > k \\ 0, & otherwise \end{cases},$$

i.e. the matrix M_k looks as

$$M_k = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -\mu_{k+1}^{(k)} & 1 & 0 & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ 0 & 0 & -\mu_n^{(k)} & 0 & 0 & 1 \end{pmatrix}.$$

Herewith, the expression for the inverse operation will be written in the form of $A^{(k-1)} = M_k^{-1} A^{(k)}$, where the matrix M_k^{-1} looks as

$$M_k^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \mu_{k+1}^{(k)} & 1 & 0 & 0 \\ \ldots & \ldots & \ldots & \ldots & \ldots & \ldots \\ 0 & 0 & \mu_n^{(k)} & 0 & 0 & 1 \end{pmatrix}.$$

As a result of the forward path of the Gaussian method,

$$A^{(n-1)} = M_{n-1} A^{(n-2)} = M_{n-1} M_{n-2} A^{(n-3)} = M_{n-1} M_{n-2} \ldots M_2 A^{(1)} = M_{n-1} M_{n-2} \ldots M_2 M_1 A^{(0)}$$ or,

vice versa, $A = A^{(0)} = M_1^{-1} A^{(1)} = M_1^{-1} M_2^{-1} A^{(2)} = M_1^{-1} M_2^{-1} \ldots M_{n-1}^{-1} A^{(n-1)}$ we shall get $A^{(n-1)}$; moreover, $A^{(n-1)} = U$ is the required upper triangular matrix, while $M_1^{-1} M_2^{-1} \ldots M_{n-1}^{-1} = L$ is the required lower triangular matrix which looks as

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \mu_2^{(1)} & 1 & 0 & 0 & 0 & 0 \\ \mu_3^{(1)} & \mu_3^{(2)} & 1 & 0 & 0 & 0 \\ \dots & \dots & \mu_{k+1}^{(k)} & 1 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ \mu_n^{(1)} & \mu_n^{(2)} & \mu_n^{(k)} & \mu_n^{(k+1)} & \dots & \mu_n^{(n-1)} & 1 \end{pmatrix}.$$

So, the desired decomposition $A = LU$ is obtained.

Let's note that the determinant of the matrix is $\det A = \det U = u_{11}u_{22}u_{33} \cdot \ldots \cdot u_{nn}$.

Such LU-decomposition of the matrix $A_{n\times n}$, once obtained (using around n^3 arithmetic operations), can then be effectively used to solve systems of linear algebraic equations of the form $Ax = b$ (as well as to find the inverse matrix A^{-1}).

Indeed, substituting the LU-decomposition into the SLAE, we obtain:

$LUx = b$, or $Ux = L^{-1}b$.

As a result, the process of solving the SLAE is reduced to two simple stages.

At the first stage, the SLAE $Lz = b$ is solved. Since the matrix of the system is lower triangular, the solution can be written in an explicit form:

$$z_1 = b_1,\ z_i = b_i - \sum_{j=1}^{i-1} l_{ij} z_j,\ i = 2,\ldots,n.$$

At the second stage, the SLAE $Ux = z$ is solved. Since the matrix of the system is upper triangular, here, as in the previous stage, the solution is presented in an explicit form:

$$x_n = \frac{z_n}{u_{nn}},\ x_i = \frac{1}{u_{ii}}(z_i - \sum_{j=i+1}^{n} u_{ij} x_j),\ i = n-1,\ldots,1.$$

Let's note that the second stage is equivalent to the backward path of the Gaussian method, while the first one corresponds to the transformation of the right-hand side of the SLAE during the forward path. Both stages require around n^2 arithmetic operations.

Finally, for inversion of the matrix A, it is necessary to solve in such a way n systems of linear equations with the same matrix $A = LU$ and different right-hand sides of the form $LUx = e_i,\ i = 1,\ldots,n$ and then construct the matrix A^{-1} from the solutions to these systems.

Example. Solve the SLAE using the Gaussian method with the help of *LU*-decomposition, calculate the determinant and the inverse matrix for the matrix of the SLAE:

$$\begin{cases} x_1 + 3x_2 - x_3 + 2x_4 = 13 \\ 6x_1 - 2x_2 + 2x_4 = 20 \\ 3x_1 - 5x_2 + x_3 + 8x_4 = 7 \\ -x_1 + 4x_2 - 5x_3 + 9x_4 = 7 \end{cases}; \; A = \begin{pmatrix} 1 & 3 & -1 & 2 \\ 6 & -2 & 0 & 2 \\ 3 & -5 & 1 & 8 \\ -1 & 4 & -5 & 9 \end{pmatrix}, \; b = \begin{pmatrix} 13 \\ 20 \\ 7 \\ 7 \end{pmatrix}.$$

LU-decomposition of the matrix $A = A^{(0)}$:

$$M_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -6 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \; A^{(1)} = M_1 A^{(0)} = \begin{pmatrix} 1 & 3 & -1 & 2 \\ 0 & -20 & 6 & -10 \\ 0 & -14 & 4 & 2 \\ 0 & 7 & -6 & 11 \end{pmatrix}.$$

$$M_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -14/20 & 1 & 0 \\ 0 & 7/20 & 0 & 1 \end{pmatrix}, \; A^{(2)} = M_2 A^{(1)} = \begin{pmatrix} 1 & 3 & -1 & 2 \\ 0 & -20 & 6 & -10 \\ 0 & 0 & -1/5 & 9 \\ 0 & 0 & -39/10 & 15/2 \end{pmatrix}.$$

$$M_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -39/2 & 1 \end{pmatrix}, \; A^{(3)} = M_3 A^{(2)} = \begin{pmatrix} 1 & 3 & -1 & 2 \\ 0 & -20 & 6 & -10 \\ 0 & 0 & -1/5 & 9 \\ 0 & 0 & 0 & -168 \end{pmatrix} = U.$$

$$L = M_1^{-1} M_2^{-1} M_3^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 1 & 0 & 0 \\ 3 & 14/20 & 1 & 0 \\ -1 & -7/20 & 39/2 & 1 \end{pmatrix}, \; L \cdot U = A.$$

The determinant of the matrix: $\det A = \prod_{i=1}^{n} u_{ii} = 1 \cdot (-20) \cdot (-1/5) \cdot (-168) = -672$.

Solution to the SLAE $Lz = b$:

$$\begin{cases} z_1 = b_1 = 13 \\ z_2 = b_2 - l_{21} z_1 = 20 - 6 \cdot 13 = -58 \\ z_3 = b_3 - l_{31} z_1 - l_{32} z_2 = 7 - 3 \cdot 13 - \frac{7}{10} \cdot (-58) = \frac{43}{5} \\ z_4 = b_4 - l_{41} z_1 - l_{42} z_2 - l_{43} z_3 = 7 - (-1) \cdot 13 - (-\frac{7}{20}) \cdot (-58) - \frac{39}{2} \cdot \frac{43}{5} = -168 \end{cases}.$$

Solution to the SLAE $Ux = z$:

$$\begin{cases} x_4 = \dfrac{z_4}{u_{44}} = \dfrac{-168}{-168} = 1 \\ x_3 = \dfrac{1}{u_{33}}(z_3 - u_{34}x_4) = \dfrac{1}{-1/5}(43/5 - 9\cdot 1) = 2 \\ x_2 = \dfrac{1}{u_{22}}(z_2 - u_{24}x_4 - u_{23}x_3) = \dfrac{1}{-20}(-58 - (-10)\cdot 1 - 6\cdot 2) = 3 \\ x_1 = \dfrac{1}{u_{11}}(z_1 - u_{14}x_4 - u_{13}x_3 - u_{12}x_2) = \dfrac{1}{1}(13 - 2\cdot 1 - (-1)\cdot 2 - 3\cdot 3) = 4 \end{cases}.$$

Solution to the SLAE: $x = \begin{pmatrix} 4 & 3 & 2 & 1 \end{pmatrix}^T$.

Finding the inverse matrix A^{-1}.

Solution to the SLAE $Lz_1 = e_1$: $z_1 = \begin{pmatrix} 1 & -6 & 6/5 & -49/2 \end{pmatrix}^T$.

Solution to the SLAE $Ux_1 = z_1$: $x_1 = \begin{pmatrix} 1/12 & 19/48 & 9/16 & 7/48 \end{pmatrix}^T$.

Solution to the SLAE $Lz_2 = e_2$: $z_2 = \begin{pmatrix} 0 & 1 & -7/10 & 14 \end{pmatrix}^T$.

Solution to the SLAE $Ux_2 = z_2$: $x_2 = \begin{pmatrix} 1/6 & -1/12 & -1/4 & -1/12 \end{pmatrix}^T$.

Solution to the SLAE $Lz_3 = e_3$: $z_3 = \begin{pmatrix} 0 & 0 & 1 & -39/2 \end{pmatrix}^T$.

Solution to the SLAE $Ux_3 = z_3$: $x_3 = \begin{pmatrix} -1/28 & 1/112 & 25/112 & 13/112 \end{pmatrix}^T$.

Solution to the SLAE $Lz_4 = e_4$: $z_4 = \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix}^T$.

Solution to the SLAE $Ux_4 = z_4$: $x_4 = \begin{pmatrix} -1/42 & -13/168 & -15/56 & -1/168 \end{pmatrix}^T$.

From here $A^{-1} = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \end{pmatrix} = \begin{pmatrix} 1/12 & 1/6 & -1/28 & -1/42 \\ 19/48 & -1/12 & 1/112 & -13/168 \\ 9/16 & -1/4 & 25/112 & -15/56 \\ 7/48 & -1/12 & 13/112 & -1/168 \end{pmatrix}$.

2.10. *LUP*-DECOMPOSITION OF MATRICES

The Gaussian method with pivot element selection is implemented on a computer with the help of *LUP*-decomposition of matrices, which is the following extension of *LU*-decomposition of the matrix *A*:

$PA = LU$,

where L is the lower triangular matrix ($l_{ij} = 0$, if $i < j$; $l_{ii} = 1$, $i = 1,\ldots,n$), U is the upper triangular matrix ($u_{ij} = 0$, if $i > j$), and P is a permutation matrix, i.e. a unity matrix with rows repeatedly swapped with each other.

The desired *LUP*-decomposition, unlike the *LU*-decomposition, works for any nondegenerate matrix and can be constructed using the Gaussian method with pivot element selection described above. Herewith, for ease of calculation and storage of the L and U matrices, a mixed matrix $C = L + U - E$ is often used instead, i.e. $c_{ij} = u_{ij}$, if $i \le j$ and $c_{ij} = l_{ij}$, if $i > j$.

To obtain the matrix $C = L + U - E$, $k = 1,\ldots,n-1$ steps are required; the initial matrices are $C^{(0)} = A$, $P^{(0)} = E$. First, at the k-th step of the method, in the matrix $C^{(k-1)}$ the pivot (maximum in modulus) element $c_{ik}^{(k-1)}$, $i = k,\ldots,n$ is selected and the corresponding permutation of rows i and k in it and in the matrix $P^{(k-1)}$ is performed, which guarantees $c_{kk}^{(k-1)} \neq 0$. Then the common part of all Gaussian methods is performed. All elements of the current k-th column located below the k-th row are divided by the pivot one:

$$c_{ik}^{(k-1)} = \frac{c_{ik}^{(k-1)}}{c_{kk}^{(k-1)}},\ i = k+1,\ldots,n,$$

after which from all elements of the matrix $C^{(k-1)}$ located below the k-th row and k-th column (i.e. $i > k$ and $j > k$) this product is subtracted:

$$c_{ij}^{(k-1)} = c_{ij}^{(k-1)} - c_{ik}^{(k-1)} \cdot c_{kj}^{(k-1)},\ i = k+1,\ldots,n,\ j = k+1,\ldots,n.$$

After all the steps of the forward path of the Gaussian method with pivot element selection have been completed, the final matrix $C = C^{(n-1)}$ can be divided into the matrices L and U. As a result, taking into account the permutation matrix $P = P^{(n-1)}$, the desired decomposition $PA = LU$ is obtained.

Herewith the determinant of the matrix A:

$$\det A = \det P \cdot \det U = (-1)^p \cdot u_{11}u_{22}u_{33} \cdot \ldots \cdot u_{nn},$$

where p is the amount of row permutations performed.

The resulting *LUP*-decomposition of the matrix $A_{n\times n}$ can be effectively used to solve systems of linear algebraic equations of the form $Ax=b$ and to find the inverse matrix A^{-1}.

Indeed, substituting the *LUP*-decomposition into the SLAE, we obtain:

$PAx=Pb$, $LUx=Pb$, or $Ux=L^{-1}Pb$.

The process of solving the SLAE is again reduced to two stages, at each of which the solution can be written in an explicit form due to the triangular matrix.

At the first stage, the SLAE $Lz=Pb=d$ is solved:

$$z_1=d_1,\ z_i=d_i-\sum_{j=1}^{i-1}l_{ij}z_j,\ i=2,\ldots,n.$$

At the second stage, the SLAE $Ux=z$ is solved:

$$x_n=\frac{z_n}{u_{nn}},\ x_i=\frac{1}{u_{ii}}(z_i-\sum_{j=i+1}^{n}u_{ij}x_j),\ i=n-1,\ldots,1.$$

The inversion of the matrix A is done through the solution of n systems of linear equations with the matrix $A=LU$ and different right-hand sides of the form $LUx=Pe_i$, $i=1,\ldots,n$ and the subsequent construction of the matrix A^{-1} from the solutions to these systems.

Program implementation of *LUP*-decomposition of matrices often uses, instead of the permutation matrix, a permutation vector P containing the order of the rows in the matrix in the form of corresponding indices, i.e. as

$p_{ij}=1\Leftrightarrow p_i=j$.

Then such an initial vector looks like $p_i=i, i=1,\ldots,n$.

Example. Solve the SLAE using the Gaussian method with the help of *LUP*-decomposition, calculate the determinant and the inverse matrix for the matrix of the SLAE:

$$\begin{cases}x_1+3x_2-x_3+2x_4=13\\6x_1-2x_2+2x_4=20\\3x_1-5x_2+x_3+8x_4=7\\-x_1+4x_2-5x_3+9x_4=7\end{cases};\ A=\begin{pmatrix}1&3&-1&2\\6&-2&0&2\\3&-5&1&8\\-1&4&-5&9\end{pmatrix},\ b=\begin{pmatrix}13\\20\\7\\7\end{pmatrix}.$$

LUP-decomposition of the matrix $A=C^{(0)}$, $P^{(0)}=\begin{pmatrix}1&2&3&4\end{pmatrix}^T$:

$$C^{(1)} = \begin{pmatrix} 6 & -2 & 0 & 2 \\ 1/6 & 10/3 & -1 & 5/3 \\ 1/2 & -4 & 1 & 7 \\ -1/6 & 11/3 & -5 & 28/3 \end{pmatrix},\ P^{(1)} = \begin{pmatrix} 2 & 1 & 3 & 4 \end{pmatrix}^T,\ p = 1.$$

$$C^{(2)} = \begin{pmatrix} 6 & -2 & 0 & 2 \\ 1/2 & -4 & 1 & 7 \\ 1/6 & -5/6 & -1/6 & 15/2 \\ -1/6 & -11/12 & -49/12 & 63/4 \end{pmatrix},\ P^{(2)} = \begin{pmatrix} 2 & 3 & 1 & 4 \end{pmatrix}^T,\ p = 2.$$

$$C^{(3)} = \begin{pmatrix} 6 & -2 & 0 & 2 \\ 1/2 & -4 & 1 & 7 \\ -1/6 & -11/12 & -49/12 & 63/4 \\ 1/6 & -5/6 & 2/49 & 48/7 \end{pmatrix} = C,\ P^{(3)} = \begin{pmatrix} 2 & 3 & 4 & 1 \end{pmatrix}^T = P,\ p = 3.$$

Thus, the matrices from C for the decomposition $PA = LU$ are:

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ -1/6 & -11/12 & 1 & 0 \\ 1/6 & -5/6 & 2/49 & 1 \end{pmatrix},\ U = \begin{pmatrix} 6 & -2 & 0 & 2 \\ 0 & -4 & 1 & 7 \\ 0 & 0 & -49/12 & 63/4 \\ 0 & 0 & 0 & 48/7 \end{pmatrix}.$$

It is easy to check that the product

$$L \cdot U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 \\ -1/6 & -11/12 & 1 & 0 \\ 1/6 & -5/6 & 2/49 & 1 \end{pmatrix} \cdot \begin{pmatrix} 6 & -2 & 0 & 2 \\ 0 & -4 & 1 & 7 \\ 0 & 0 & -49/12 & 63/4 \\ 0 & 0 & 0 & 48/7 \end{pmatrix} = \begin{pmatrix} 6 & -2 & 0 & 2 \\ 3 & -5 & 1 & 8 \\ -1 & 4 & -5 & 9 \\ 1 & 3 & -1 & 2 \end{pmatrix}$$

is the matrix A with rows permuted according to the rule P.

The determinant of the matrix:

$$\det A = (-1)^p \prod_{i=1}^{n} u_{ii} = (-1)^3 \cdot 6 \cdot (-4) \cdot (-49/12) \cdot (48/7) = -672.$$

Solution to the SLAE $Lz = Pb = d$, where $d_i = b_{P[i]}$:

$$\begin{cases} z_1 = d_1 = 20 \\ z_2 = d_2 - l_{21} z_1 = 7 - \frac{1}{2} \cdot 20 = -3 \\ z_3 = d_3 - l_{31} z_1 - l_{32} z_2 = 7 - (-\frac{1}{6}) \cdot 20 - (-\frac{11}{12}) \cdot (-3) = \frac{91}{12} \\ z_4 = d_4 - l_{41} z_1 - l_{42} z_2 - l_{43} z_3 = 13 - \frac{1}{6} \cdot 20 - (-\frac{5}{6}) \cdot (-3) - \frac{2}{49} \cdot \frac{91}{12} = \frac{48}{7} \end{cases}.$$

Solution to the SLAE $Ux = z$:

$$\begin{cases} x_4 = \dfrac{z_4}{u_{44}} = \dfrac{48/7}{48/7} = 1 \\ x_3 = \dfrac{1}{u_{33}}(z_3 - u_{34}x_4) = \dfrac{1}{-49/12}(91/12 - 63/4 \cdot 1) = 2 \\ x_2 = \dfrac{1}{u_{22}}(z_2 - u_{24}x_4 - u_{23}x_3) = \dfrac{1}{-4}(-3 - 7 \cdot 1 - 1 \cdot 2) = 3 \\ x_1 = \dfrac{1}{u_{11}}(z_1 - u_{14}x_4 - u_{13}x_3 - u_{12}x_2) = \dfrac{1}{6}(20 - 2 \cdot 1 - 0 \cdot 2 - (-2) \cdot 3) = 4 \end{cases}.$$

Solution to the SLAE: $x = \begin{pmatrix} 4 & 3 & 2 & 1 \end{pmatrix}^T$.

Finding the inverse matrix A^{-1}.

Solution to the SLAE $Lz_1 = Pe_1$: $z_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix}^T$.

Solution to the SLAE $Ux_1 = z_1$: $x_1 = \begin{pmatrix} 1/12 & 19/48 & 9/16 & 7/48 \end{pmatrix}^T$.

Solution to the SLAE $Lz_2 = Pe_2$: $z_2 = \begin{pmatrix} 1 & -1/2 & -7/24 & -4/7 \end{pmatrix}^T$.

Solution to the SLAE $Ux_2 = z_2$: $x_2 = \begin{pmatrix} 1/6 & -1/12 & -1/4 & -1/12 \end{pmatrix}^T$.

Solution to the SLAE $Lz_3 = Pe_3$: $z_3 = \begin{pmatrix} 0 & 1 & 11/12 & 39/49 \end{pmatrix}^T$.

Solution to the SLAE $Ux_3 = z_3$: $x_3 = \begin{pmatrix} -1/28 & 1/112 & 25/112 & 13/112 \end{pmatrix}^T$.

Solution to the SLAE $Lz_4 = Pe_4$: $z_4 = \begin{pmatrix} 0 & 0 & 1 & -2/49 \end{pmatrix}^T$.

Solution to the SLAE $Ux_4 = z_4$: $x_4 = \begin{pmatrix} -1/42 & -13/168 & -15/56 & -1/168 \end{pmatrix}^T$.

From here $A^{-1} = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \end{pmatrix} = \begin{pmatrix} 1/12 & 1/6 & -1/28 & -1/42 \\ 19/48 & -1/12 & 1/112 & -13/168 \\ 9/16 & -1/4 & 25/112 & -15/56 \\ 7/48 & -1/12 & 13/112 & -1/168 \end{pmatrix}$.

2.11. PROGRAM #04

Below is a proposed variant of the program algorithm for solving a SLAE, calculating the determinant and the inverse matrix of the SLAE's matrix using the Gaussian method with the help of *LU*-decomposition or *LUP*-decomposition.

ALGORITHM “Gaussian method with the help of *LUP*-decomposition”

INPUT n, a[n][n], b[n], l

OUTPUT i, j, k, m, p, f, g, q[n], c[n][n], d[n], x[n], y, z[n][n]

```
BEGIN
    p:=0
    CYCLE “Rows” FOR i FROM 1 TO n BY 1
        q[i]:=i
    CYCLE “LUP-decomposition” FOR k FROM 1 TO n-1 BY 1
        #Part for the LUP-decomposition
        IF (l≠0)
            IF (a[k][k]>0) g:=a[k][k] ELSE g:=-a[k][k]
            m:=k
            CYCLE “Rows” FOR i FROM k+1 TO n BY 1
                IF (a[i][k]>0) f:=a[i][k] ELSE f:=-a[i][k]
                IF (f>g) g:=f, m:=i
            IF (m≠k)
                CYCLE “Columns” FOR j FROM 1 TO n BY 1
                    g:=a[k][j], a[k][j]:=a[m][j], a[m][j]:=g
                g:=q[k], q[k]:=q[m], q[m]:=g
                p:=p+1
        #General part of the LU-decomposition
        IF (a[k][k]=0) END
        CYCLE “Rows” FOR i FROM k+1 TO n BY 1
            a[i][k]:=a[i][k]/a[k][k]
            CYCLE “Columns” FOR j FROM k+1 TO n BY 1
                a[i][j]:=a[i][j]-a[i][k]*a[k][j]
    c[1][1]:=1
    CYCLE “Rows” FOR i FROM 2 TO n BY 1
        CYCLE “Columns” FOR j FROM 1 TO i-1 BY 1
            c[i][j]:=a[i][j]
            a[i][j]:=0
        c[i][i]:=1
    y:=1-2*REMAINDER(p,2)
```

```
        CYCLE “Determinant” FOR i FROM 1 TO n BY 1
            y:=y*a[i][i]
        CYCLE “First stage” FOR i FROM 1 TO n BY 1
            d[i]:=b[q[i]]
            CYCLE “Item” FOR j FROM 1 TO i-1 BY 1
                d[i]:=d[i]-c[i][j]*d[j]
        CYCLE “Second stage” FOR i FROM n TO 1 BY -1
            x[i]:=d[i]
            CYCLE “Item” FOR j FROM i+1 TO n BY 1
                x[i]:=x[i]-a[i][j]*x[j]
            x[i]:=x[i]/a[i][i]
        CYCLE “Inverse matrix” FOR k FROM 1 TO n BY 1
            CYCLE “Rows” FOR i FROM 1 TO n BY 1
                b[i]:=0
            b[k]:=1
            CYCLE “First stage” FOR i FROM 1 TO n BY 1
                d[i]:=b[q[i]]
                CYCLE “Item” FOR j FROM 1 TO i-1 BY 1
                    d[i]:=d[i]-c[i][j]*d[j]
            CYCLE “Second stage” FOR i FROM n TO 1 BY -1
                z[i][k]:=d[i]
                CYCLE “Item” FOR j FROM i+1 TO n BY 1
                    z[i][k]:=z[i][k]-a[i][j]*z[j][k]
                z[i][k]:=z[i][k]/a[i][i]
        PRINT x, y, z
END
```